



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
AUGS DIVISION

First Semester 2026-27

Course Handout Part II

Date: 1 August 2026

In addition to Part-I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number :CHEM F335

Course Title :Organic Chemistry and Drug Design

Instructor-in-charge :Dalip Kumar (email- dalipk@pilani.bits-pilani.ac.in, chamber no.3268-T)

- Course description:** An introduction to organic chemistry principles and reactivities vital to drug design, drug development and drug action; the role of molecular size, shape, and charge, and in drug action; proteins and nucleic acids as drug targets; bioisosterism; ADME, QSAR and drug design; applied molecular modeling and combinatorial synthesis; Synthesis of some selected chemotherapeutic agents (antibacterial, antiviral agents, antimalarial, anticancer etc.)
- Scope and objective:** To familiarize the students with basic aspects of drug discovery and more importantly, the applications of organic chemistry in drug design, important drug targets, marketed drugs, synthesis of drugs; the overall objective is to have a reflective teaching and learning environment
- Course learning outcomes:** The student should be able to accomplish the following:
CL01. Understanding about structure & function of receptors, enzymes, and nucleic acids
CL02. Able to identify and list at least five biological targets for drugs
CL03. For the above targets be able to identify ten important available drugs with molecular mechanism
CL04. Be able to use computational software/tools to determine given properties of the drug target of his choice
CL05. At the end, be able to rationally design and propose simple synthesis of a drug for a disease.

- Text Book (TB1):** An Introduction to Medicinal Chemistry by Graham L. Patrick, Oxford University Press 5th Ed.

Reference Books (RB1): The organic chemistry of drug design and drug action 2nd edition
by Richard B Silverman.

5. Course Format:

Module no.	Description	Ref.	Learning outcome
1-6	Drugs and Drug targets; Intermolecular bonding forces. Structure and function of protein, enzyme, receptors and nucleic acids	TB1: Ch.1-4, 6 & 7	general definition of a drug, amino acids, enzymes, receptors & nucleic acids (structure and function) drug targets & molecular interactions
7-10	Enzyme inhibitors; Receptor role and action; affinity, efficacy, and potency. Basic idea of PK; its applications to drug discovery; elementary idea of Absorption, Distribution, Metabolism, Elimination, and Toxicity (ADMET)	TB1: Ch. 7, 11	general mechanisms of drugs to produce a pharmacological or biological effect, drug absorption, metabolism and excretion
11-14	Drug discovery, design, and development choosing a disease, target, and bioassay; finding leads from natural drugs, libraries, existing drugs, optimizing target interactions & access to the target	TB1: Ch. 12-14	design of a chemical entity, structural modifications, optimization of activity



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15-19	Physicochemical properties; Hansch equation; the Craig plot; the Topliss scheme; Bioisosteres; the Free-Wilson approach; planning a QSAR study; 3-D QSAR; case study.	TB1: Ch. 18	roles of hydrophobicity, electronic and steric parameters in drug design, tools to study QSAR
20-24	Sulfonamides; penicillins; cephalosporins; β -lactam antibiotics: introduction, synthesis, and design	TB1: Ch. 19	mechanism of action, types, synthesis and structural modification of antibacterial agents
25-29	Against DNA viruses; against RNA viruses/HIV; protease inhibitors: introduction; synthesis, and design	TB1: Ch. 20	structure and life cycle of viruses, types of antiviral drugs their synthesis, structure and mode of actions
30-35	Cancer; drugs acting on nucleic acids, structural proteins and enzymes; Photodynamic Therapy	TB1: Ch. 21	causes of cancer, types of anticancer agents and their mechanism of action, structures and syntheses of selected anticancer drugs
36-40	antimalarial drugs (quinine, chloroquine, mefloquine and artemisinin) analgesics, antipyretics	TB1: Ch. 24 & case study page 300	structure, synthesis and mode of actions of selected antimalarial, analgesic and antipyretic drugs

6. Evaluation Scheme:

Component	Weightage (%)	Duration	Date and Time	Comments
Continuous Evaluation	30	TBA*	TBA*	tests /Quiz/assignment
Mid-semester test	30	90 min	According to the timetable	close book
Comprehensive Examination	40	3 h	According to the timetable	partly close book & partly open book

*TBA: To Be Announced

- 7. Minimum criterion for awarding valid grades:** Student should obtain 30% of the average of the top five performers in the class, or 40% of the median marks of the class, whichever is lower to clear the course. If any student gets the marks lower than the prescribed standard mentioned above, he/she may be awarded NC.
- 8. Chamber consultation hours:** Friday 12.00-01 PM, Room No 3268T
- 9. Makeup Policy:** Makeup will be given only for genuine cases (accompanied by proof) for Mid-sem or comprehensive exam only.
- 10. Attendance Policy:** For continuous evaluation linked to attendance, final grades will be awarded only to students who maintain a minimum attendance of 50%.
- 11. Notices:** All the notices pertaining to this course will be displayed on **Nalanda Website**.

Instructor-in-Charge